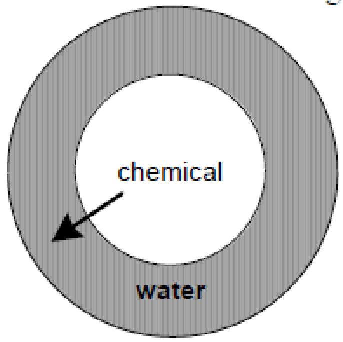
P6.32

In the inner chamber, the chemical is combusted. Heat, as indicated by the arrow, travels between the inner and outer chamber. No mass is exchanged between the two chambers.



First, let's consider the water as the system. We use an integral energy balance equation, with no mass entering or leaving the system, no change in kinetic or potential energy, and no work:

$$U_{sys,f} - U_{sys,0} = Q$$

The change in energy is at constant volume and is due to a change in temperature. For water, $C_p = C_v = 4.18$ J/g °C, and

$$U_{sys,f} - U_{sys,0} = Q = mC_v\left(T_f - T_0\right) = 650 \times 4.18 \times (36.2 - 23.6) = 34234 \text{ J}$$

Now we consider the inner chamber as the system. We have

$$U_{sys,f} - U_{sys,0} = Q = -34234 \text{ J}$$

A chemical reaction takes place in the inner chamber:

$$C_8H_{10}O_3 + 9O_2 \rightarrow 8CO_2 + 5H_2O$$

As a reasonable approximation, we assume that virtually all of the change in energy of this system is due to the reaction (we neglect the small increase in temperature). Therefore

$$U_{sys,f} - U_{sys,0} = n_r \Delta\hat{U}_r = -34234 \text{ J}$$

where n_r is the moles of reaction, which is 2.7 g/154 g/gmol or 0.0175 gmol. Therefore,

$$\Delta\hat{U}_r = -\frac{34234 \text{ J}}{0.0175 \text{ gmol}} = -1956 \text{ kJ/gmol}$$

There is a change in moles in the gas phase, if we assume the chemical and water are in the liquid phase then this change is (9 – 8) or -1 mole, therefore

$$\Delta\hat{H}_r = \Delta\hat{U}_r + RT\Delta n_R = -1956 + \left(0.0083144 \times 298 \times (-1)\right) = -1958 \text{ kJ/gmol}$$

Now we know that

$$\Delta\hat{H}_r = \sum v_i \Delta\hat{H}_{f,i} = (-1)\Delta\hat{H}_{f,C_8H_{10}O_3} + (-9)(0) + 8(-393.5) + 5(-285.84) = -1958 \text{ kJ/gmol}$$

or

$$\Delta\hat{H}_{f,C_8H_{10}O_3} = -2620 \text{ kJ/gmol}$$

P6.34

The reaction is

$$2\text{Fe} + \frac{3}{2}\text{O}_2 \rightarrow \text{Fe}_2\text{O}_3$$

which is just the formation reaction for Fe_2O_3, for which

$$\Delta\hat{H}_f = -830.5 \text{ kJ/gmol}$$

The total heat release desired is 600 kJ. (How would you design the hot pack to control the rate of heat release??) Therefore we need to generate (-600/-830.5) or 0.722 gmol Fe_2O_3, which requires that 1.44 gmol Fe react. Since the molar mass of Fe is 56 g/gmol, we should put (1.44)(56) or about 80 g Fe filings in the hot pack.

Oxygen is supplied by diffusion into the hotpack from the air. The final mass is 0.722 gmol x 160 g/gmol or 115 g Fe_2O_3. The mass increases about 35 g, or by about 44%.

P6.37

The reaction is

$$\mathrm{Na} + \mathrm{H_2O} \rightarrow \frac{1}{2}\mathrm{H_2} + \mathrm{NaOH}$$

$$\Delta \hat{H}_r^\circ = \sum_i \nu_{i1} \Delta \hat{H}_{f,i}^\circ = (-1)(-285.84) + 1(-425.9) = -140 \text{ kJ/gmol}.$$

1 mL water = 1 g water = 0.056 gmol. 100 g Na = 4.35 gmoles. If all water reacts, about 4.35 – 0.056 = 4.29 gmol or about 98.7 g of Na remain. The heat capacity of Na is 28 J/gmol °C (Lange's Handbook of Chemistry). The energy balance (assuming no heat and no work) is

$$(-140 \text{ kJ/gmol reaction}) \times (0.056 \text{ gmol}) + (4.29 \text{ gmol}) \times (28 \text{ J.gmol °C}) \times (\Delta T) = 0$$

$$\Delta T = 65°C$$

The heat released upon reaction is sufficient to warm the Na metal up by about 65°C, or, to about 85 - 90°C from room temperature. This may be enough to cause the hydrogen gas to explode.

If a small amount of sodium metal added to a large quantity of water, the high heat capacity of the water tends to "absorb" the heat of reaction – the water temperature will rise, but the hydrogen gas won't ignite.